BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ELECTRONICS MATERIALS & INSTRUMENTATION (2022)

Exam Date: Saturday, 7 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2022	INSTRUMENTATION AND CONTROL	102404	B.Tech 4th Semester Exam., 2022

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Exam., 2022

INSTRUMENTATION AND CONTROL

Time: 3 hours | Code: 102404 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[2x7=14]**

1. Which of the following causes a controlled variable to change?
   i. Disturbance
   ii. Change in set point
   iii. Variation in load demand
   iv. All of the above

2. What is the resolution of a DAC with a maximum voltage of 15 volts and five input lines?
   i. 0.4687
   ii. 0.4838
   iii. 0.2343
   iv. 0.2419

3. The derivative function gives the actuator _____.
   i. boost action
   ii. braking action
   iii. Both (i) and (ii)
   iv. None of the above

4. A square wave that is 20 volts at its On state and 0 volt at its Off state will produce an average DC voltage of _____ when duty cycle is 75%.
   i. 15 V
   ii. 10 V
   iii. 7.5 V
   iv. 5 V

5. The resistance of an RTD is:
   i. directly proportional to the temperature
   ii. inversely proportional to the temperature
   iii. directly proportional to the square of temperature
   iv. inversely proportional to the square of temperature

6. Which of the following valves has the greatest flow capacity of all control valves?
   i. Single-seated
   ii. Three-way
   iii. Butterfly
   iv. Ball

7. A unity feedback system has open-loop transfer function as $G(s) = \frac{16}{s(s+16)}$. The natural frequency of the system is:
   i. 16
   ii. 8
   iii. 2
   iv. 4

8. Which of the following represents the application of inductive transducers?
   i. Displacement measurement
   ii. Thickness measurement
   iii. Both displacement and thickness measurements
   iv. None of the above

9. Which of the following can act as a comparator?
   i. Op-amp with negative feedback
   ii. Op-amp with positive feedback
   iii. Op-amp without feedback
   iv. None of the above

10. Transfer function represents the input-output relationship of a system, with initial conditions as:
    i. zero
    ii. positive
    iii. negative
    iv. All of the above

Q.2 Solve both questions : **[14]**

1. Define the following: (i) Transducer (ii) Resolution (iii) Linearity (iv) Accuracy (v) Repeatability (vi) Calibration (vii) Backlash

2. Discuss in detail about the classification and elimination of error.

Q.3 Solve both questions : **[14]**

1. Differentiate among piezoelectric, inductive and capacitance-type transducers.

2. What are pyrometers? Explain with the help of neat sketch.

Q.4 Solve both questions : **[14]**

1. Explain the working of electrical resistance strain gauge. Explain how the resistance change of a strain gauge is measured with the help of Wheatstone bridge. Also, explain the methods of lead wire compensation and temperature compensation in case of strain gauge.

2. Explain the five basic laws of thermocouple behaviour and how these laws are useful in designing thermocouple for temperature measurement.

Q.5 Solve all questions : **[14]**

1. Consider a system with controller G_c, reference input R, and disturbance input D acting on the plant $\frac{1}{s(s+2)}$. If G_c is a proportional controller, $G_c = K_p$, what value of K_p will cause the steady-state error for a unit ramp reference input to be 0.1?

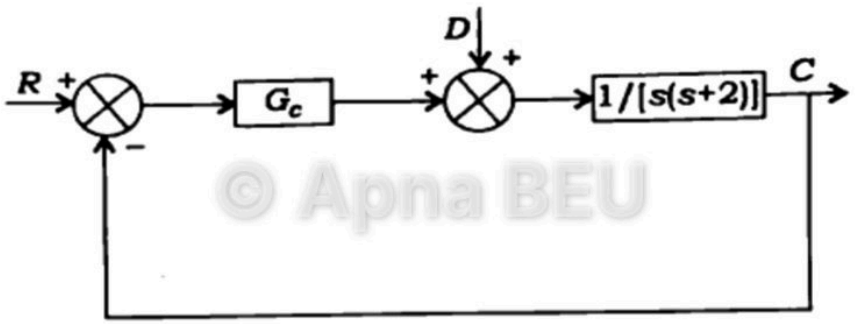
   (a) If G_c is a proportional controller, $G_c = K_p$, what value of K_p will cause the steady-state error for a unit ramp reference input to be 0.1?

   (b) If zero steady-state error for a unit ramp reference input is required, what form should the controller G_c take?

   (c) For a unit-step disturbance input, what form of controller G_c should be chosen to make the steady-state error zero?

   (d) Assume that the reference input is a unit step, the disturbance input is zero and proportional controller being used. What value (or values) of K_p will yield an underdamped system with a settling time of 4 seconds?

   (e) Considering the same case as in part (d), what effect does increasing K_p have on the damping ratio?



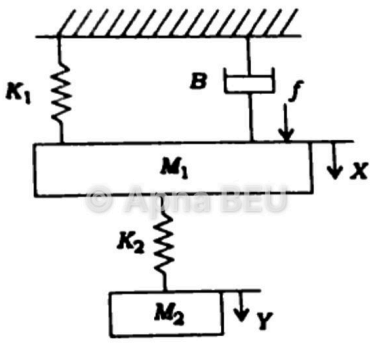
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Figure 1

Q.6 Solve both questions : **[14]**

1. With the help of a suitable example, explain the difference between open-loop control and closed-loop control.

2. For a given mechanical translational system with masses M_1, M_2, springs K_1, K_2, and damper B, find out the transfer function $\frac{X(s)}{F(s)}$.



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Figure 2

Q.7 Solve both questions : **[14]**

1. Explain the operations of the following: (i) Inverting op-amp (ii) Summing op-amp (iii) Differentiating op-amp (iv) Integrating op-amp.

2. What are the important requirements of a signal conditioner and how are they met?

Q.8 Solve this question : **[14]**

1. For the open-loop transfer function $G(s) = \frac{400(s+1)(s+10)}{s^3(s+100)(s+1000)}$, sketch the Bode magnitude and phase diagrams. Sketch the diagrams for each term and then also sketch the total diagrams. Show the slopes of each section and of the total diagram.

Q.9 Write short notes on any four of the following: **[14]**

1. a. Sensitivity analysis
   b. Pneumatic actuators
   c. Flow measurement sensors
   d. ADC using successive-approximation register (SAR)
   e. Optical encoders